

Computational Spatiotemporal Framework for Special Event Traffic Forecasting and Resource Deployment Optimization

Operational Challenges and the Breakdown of Traditional Traffic Management

Urban transportation networks are highly susceptible to severe localized bottlenecks and sudden cascade failures triggered by planned special events and unplanned disruptions.¹ Planned special events, such as high-attendance sports fixtures, major political rallies, community religious processions, and massive infrastructure construction projects, introduce sharp, localized surges in traffic demand.³ Conversely, unplanned incidents—including sudden waterlogging from localized storms, vehicle breakdowns on high-priority corridors, tree falls, and multi-vehicle collisions—abruptly constrain available physical capacity.⁵ Traditional traffic incident management and traffic planning methodologies remain fundamentally reactive.⁷ Municipalities and local traffic enforcement agencies face three critical operational bottlenecks:

- **Absence of Pre-Event Impact Quantification:** Traffic management centers lack the mathematical models needed to simulate and quantify the network-wide spatiotemporal footprint of a planned special event before its occurrence.³ Consequently, the geographic boundary, queue propagation patterns, and expected travel delays are left unanalyzed.⁴
- **Experience-Driven, Subjective Resource Allocation:** The deployment of traffic enforcement personnel, physical barricades, and corridor-level diversions is driven by historical precedent and the subjective decisions of local patrol administrators.¹¹ This unstructured approach ignores complex spatiotemporal dependencies and fails to optimize personnel shifts, leading to over-staffing at certain posts and critical resource gaps at active bottleneck junctions.¹³
- **Lack of Post-Event Learning Systems:** Municipalities do not maintain systematic after-action pipelines to compare plan-versus-actual traffic metrics.⁸ Lessons learned from operational failures are not codified, which prevents predictive models and resource allocation algorithms from adapting to changing urban dynamics over time.¹⁰

A systematic analysis of historical traffic logs reveals the diversity of these localized disruptions. The table below represents a categorized profile of urban events and incident-driven breakdowns:

Event Identifier	Target Corridor Class	Event Classification	Geographical Coordinates	Localized Address / Landmark	Underlying Event Cause	Operational State & Comments
FKID000008 ⁵	CBD 2 Corridor (High Priority)	Planned Event	12.97883573, 77.59953728	Queens Statue Circle near M. Chinnaswamy Stadium	Cricket Match	Closed; high-density spectator congestion managed near the stadium. ⁵
FKID000502 ⁵	ORR East 1 (High Priority)	Planned Event	12.92609339, 77.67623685	Outer Ring Road, Bellandur Intersection	Palakki Uthsava Festival	Closed; multi-point route path polygon tracked via coordinate sets. ⁵
FKID001734 ⁵	Non-Corridor (Low Priority)	Planned Event	13.01495345, 77.65422187	Chikka Banaswadi, Bengaluru East	Brahma Rathotsava & Palanquin Festival	Closed; massive localized procession requiring complete road closure. ⁵
FKID001896 ⁵	Non-Corridor (Low Priority)	Planned Event	12.88601305, 77.50546757	NICE Expressway, Banashankari 6th	VIP Movement (PM Visit)	Closed; strict road restrictions and

				Stage		dynamic security closures enforced. ⁵
FKID000010 ⁵	ORR East 2 (High Priority)	Unplanned Incident	13.00084570, 77.68137120	Whitefield Road, ITI Data Center	Severe Waterlogging	Closed; storm drainage overflow causing slow vehicle movement. ⁵
FKID000027 ⁵	Bannerghatta Road (High Priority)	Unplanned Incident	12.90773830, 77.60057090	JD Mara Junction, BTM Layout Stage 2	Heavy Vehicle Breakdown (Bus)	Closed; BMTC bus experienced gearbox failure, blocking the high-volume left lane. ⁵
FKID0000201 ⁵	CBD 2 Corridor (High Priority)	Unplanned Incident	12.98699590, 77.57233410	Vidhana Veedhi, Shivaji Nagar	Damaged Road Conditions	Closed; collapsed manhole cover and sewage overflow causing severe delays. ⁵
FKID004085 ⁵	Varthur Road	Planned Event	12.95691291,	Marathahalli Bridge	Metro Infrastructure	Closed; service

	(High Priority)		77.70135409	Service Road	ture Construction	road construction narrowing physical capacity to a single lane. ⁵
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These heterogeneous disruptions demonstrate that planned special events impose a structural demand surge on the road network, while unplanned incidents abruptly reduce active physical capacity.⁴ Transitioning from manual, subjective deployment to a predictive spatiotemporal system requires the convergence of predictive traffic forecasting, multi-objective resource optimization, and a closed-loop system for post-event learning.⁷

Spatiotemporal Forecasting and Multi-Modal Traffic Flow Prediction

Predicting urban traffic flow and congestion patterns under event-driven demand requires modeling both spatial dependencies and temporal dynamics.¹⁸ Modern architectures integrate Graph Neural Networks (GNNs) with temporal deep learning models, such as Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) networks.¹

Graph Representation of Road Networks

The transportation network is modeled as a directed, weighted graph $G = (V, E, w)$.²²

The set of vertices V represents physical street intersections and endpoints.²² The set of directed edges $E \subseteq V \times V$ represents road segments connecting these intersections.²²

The dynamic weight function $w : E \times T \rightarrow \mathbb{R}^+$ assigns a positive cost representing the estimated travel time or congestion level on edge $e \in E$ at a specific time step $t \in T$.²² This weight is computed based on physical characteristics, such as the segment length, number of active lanes, and real-time vehicle density²²:

$$\text{Density } (D_{e,t}) = \frac{Q_{e,t}}{v_{e,t}}$$

where $Q_{e,t}$ represents the traffic flow volume (vehicles per hour) passing a fixed detection sensor on edge e during time interval t , and $v_{e,t}$ denotes the average vehicle speed on that segment.²⁴

Hybrid Deep Learning Architectures

While GNNs capture non-Euclidean spatial dependencies by propagating traffic state information among adjacent and downstream road segments, recurrent structures capture the temporal evolution of these states.²⁰ The bidirectional Long Short-Term Memory (BiLSTM) network processes sequential traffic data in both forward and backward temporal directions.²⁵ This symmetric extraction capability enables the model to simultaneously capture historical dependencies and future downstream propagation trends.²⁵

To improve forecasting performance under dynamic conditions, the system integrates advanced hybrid models:

- **DTSA-3GCN (Decomposed Temporal Self-Attention Multi-layer Graph Convolutional Network):** This architecture combines Singular Value Decomposition (SVD) and Self-Attention mechanisms with a temporal multi-layer GCN to prevent long-term prediction degradation, maintaining forecasting stability across extended horizons.²⁷
- **Wavelet Denoising + XGBoost:** High-frequency noise caused by random traffic fluctuations is removed using the Mallat multi-scale wavelet decomposition algorithm.²⁸ The reconstructed single-branch sequences are then processed by an Extreme Gradient Boosting (XGBoost) model optimized via Bayesian algorithms, enabling highly accurate short-term prediction of volume surges and travel times.²⁸

Integration of Heterogeneous Contextual Features

To accurately forecast event-driven anomalies, the model incorporates non-traffic external data directly into the feature space.¹⁸ The spatiotemporal graph neural network integrates:

- **Event Schedule and Attendance Profiles:** Anticipated crowd size, event classification, ingress/egress profiles, and historical ticket scan rates.³¹
- **Dynamic Social Media Parsing:** Real-time extraction of unstructured text feeds (e.g., citizen alerts, municipal notices) processed via a Word2Vec-BiGRU-CNN architecture to classify incident types and identify early-stage bottlenecks.¹⁷
- **Meteorological Data:** Real-time and forecasted weather parameters (e.g., rainfall intensity, temperature, visibility) that scale the maximum capacity of local segments.²⁰
- **Non-Homogeneous Segment Density:** Unlike traditional models that treat density uniformly, the system models segment-specific density variations, recognizing that localized events impact adjacent lanes and intersections unevenly.¹⁷

Traffic Incident Duration and Clearance Time Predictive Analytics

When unplanned incidents occur during a special event, the traffic management center must predict the incident clearance duration to assess its impact on the network.³³ This forecasting task is divided into four distinct operational phases:

$$\text{Total Incident Duration} = T_{\text{detection}} + T_{\text{response}} + T_{\text{clearance}} + T_{\text{recovery}}$$

where $T_{\text{detection}}$ is the time from incident occurrence to reporting, T_{response} is the time elapsed until the physical arrival of emergency response teams, $T_{\text{clearance}}$ is the active time taken to remove debris and clear travel lanes, and T_{recovery} is the time required for the affected segments to return to free-flow conditions.⁶

Survival Analysis and Hazard-Based Duration Modeling

Because the clearance process represents a time-to-event survival scenario, statistical and machine learning models are developed within a survival analysis framework to capture duration dependence.³⁵ The probability of an incident clearing at time step t , given that it has lasted up to t , is modeled using the hazard function $\lambda(t)$ ³⁵:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

The corresponding survival function $S(t)$, representing the probability that the incident remains uncleared beyond time t , is defined as³⁶:

$$S(t) = \Pr(T > t) = \exp \left(- \int_0^t \lambda(u) du \right)$$

Advanced deep survival networks, such as DeepSurv and Multi-Task Deep Neural Networks (MTDNN), combine deep learning with non-parametric hazard estimation.³⁵ The MTDNN simultaneously estimates the mean clearance time and the overall survival curve, sharing learned spatiotemporal features to improve prediction accuracy.³⁵ Rather than outputting a single deterministic value, these models output a continuous survival curve for each active incident.³⁶ The median survival time—the point where $S(t) = 0.50$ —is utilized as the

operational prediction target.³⁶

Empirical Modeling and Validation

Empirical evaluations using real-world expressway incident databases demonstrate the impact of diverse predictive variables.³⁴ Sensitivity analyses using Shapley Additive Explanations (SHAP) reveal critical differences in clearance dynamics³⁴:

- **Minor Incidents:** Vehicle type, minor collisions, and mechanical breakdowns are key predictors of short clearance times, where vehicle damage severity is the primary driver.³⁹
- **Major Incidents:** Crash severity, heavy vehicle involvement (e.g., freight trucks, buses), and hazardous material spills dominate long-clearance-time predictions.³⁴ These incidents require specialized towing equipment, extended medical response, and multi-agency coordination.³⁴

A comparative analysis of the primary forecasting and clearance duration models is summarized below:

Predictive Model Type	Algorithmic Mechanism	Data Inputs & Feature Matrix	Evaluation Metrics	Operational Strengths	Limitations
XGBoost ³⁷	Regularized gradient boosted decision trees. ³⁷	Temporal context, incident type, lane block count, weather, vehicle class. ³⁸	MAE, RMSE, R^2 . ³⁷	High accuracy ($\approx$); native missing data handling. ³⁷	Struggles with non-linear spatial dependencies. ²⁵
LightGBM ³⁷	Leaf-wise tree growth with gradient-based sampling. ³⁷	Large-scale categorical traffic logs, road class, spatial coordinates. ³⁷	MAE, RMSE, R^2 . ³⁷	Fast training; highly memory efficient. ³⁷	Susceptible to overfitting on small datasets. ³⁷
DeepSurv ³⁶	Feedforward neural	Spatial risk index,	Concordance Index,	Outputs continuous	Requires intensive

	network combined with Cox proportional hazards. ³⁶	collision severity, vehicle type, response dispatch lag. ³⁵	MAE, RMSE. ³⁶	survival curves and hazard probabilities. ³⁶	computing resources. ³⁶
MTDNN ³⁵	Multi-task deep neural network sharing latent representations. ³⁵	Traffic volume, geometric segment parameters, weather, responder arrival times. ³⁵	MAPE, survival probability accuracy. ³⁵	Captures duration dependence; outperforms standard hazard models by 29.7% . ³⁵	Complex hyperparameter tuning. ³⁵
Word2Vec-BiGRU-CNN ³²	Word embeddings with bidirectional GRUs and convolutional layers. ³²	Unstructured incident dispatcher text, citizen logs, emergency call notes. ³²	F1-Score ($\approx$), classification accuracy. ³²	Converts unstructured textual context into structured features. ³²	High training complexity; requires dense text corpora. ³²

Multi-Objective Mathematical Models for Manpower and Tactical Resource Deployment

Deploying traffic personnel, positioning physical barricades, and assigning patrol routes during a planned special event or an unplanned capacity reduction can be formulated as a Multi-Objective Linear Programming (MOLP) problem.¹² The deployment framework must balance operating costs, risk mitigation, and structural resource constraints.⁴¹

The model represents the road network as segments $i \in \{1, 2, \dots, m\}$ and divides the operating day into shifts $j \in \{1, 2, \dots, n\}$.⁴¹ Five distinct categories of personnel are modeled: Assistant Sub-Inspectors (a), Sergeants (s), Constables (c), Home-Guards (h), and Civic-Volunteers (v).⁴¹ The binary decision variable $x_{ijkt}^t \in \{0, 1\}$ represents whether individual k_t of personnel type $t \in \{a, s, c, h, v\}$ is deployed to road segment i during

shift j .⁴¹

The mathematical formulation utilizes four competing objective functions:

First Objective (G_1): Minimize Total Operational Cost

This objective minimizes the total wage, administrative, and logistical cost of deploying personnel across all shifts and road segments:

$$\text{Min } G_1 = \sum_{i=1}^m \sum_{j=1}^n \sum_{k_a=1}^{N_a} \text{Cost}_a x_{ijk_a}^a + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_s=1}^{N_s} \text{Cost}_s x_{ijk_s}^s + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_c=1}^{N_c} \text{Cost}_c x_{ijk_c}^c +$$

Second Objective (G_2): Maximize Sergeant Allocation on Accident-Prone Segments

This objective ensures that higher-ranking, experienced supervisory officers (Sergeants) are prioritized at high-risk, accident-prone segments, where p_{ij} is the segment risk factor and Ln_i is the segment length:

$$\text{Max } G_2 = \sum_{i=1}^m \sum_{j=1}^4 \sum_{k_s=1}^{N_s} p_{ij} \text{Ln}_i x_{ijk_s}^s$$

Third Objective (G_3): Minimize Reliance on Volunteer Personnel

This objective minimizes the deployment of temporary, non-sworn personnel (Home-Guards and Civic-Volunteers) to ensure that the primary sworn force is utilized efficiently before relying on auxiliary staff:

$$\text{Min } G_3 = \sum_{i=1}^m \sum_{j=1}^n \sum_{k_h=1}^{N_h} x_{ijk_h}^h + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_v=1}^{N_v} x_{ijk_v}^v$$

Fourth Objective (G_4): Maximize Direct Regulatory Contacts

This objective maximizes the active enforcement capability (such as checking for emission compliance and traffic violations) of the deployed staff, where RV_t is the enforcement rate factor associated with personnel type t :

$$\text{Max } G_4 = \sum_{i=1}^m \sum_{j=1}^n \sum_{k_a=1}^{N_a} RV_a x_{ijk_a}^a + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_s=1}^{N_s} RV_s x_{ijk_s}^s + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_c=1}^{N_c} RV_c x_{ijk_c}^c + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_h=1}^{N_h} RV_h x_{ijk_h}^h + \sum_{i=1}^m \sum_{j=1}^n \sum_{k_v=1}^{N_v} RV_v x_{ijk_v}^v$$

The optimization is subject to the following functional constraints:

- **Shift Limitations:** To protect officer wellness and comply with labor standards, no individual officer of any personnel type t can be scheduled for more than two shifts in a single day:

$$\sum_{j=1}^4 x_{ijk_t}^t \leq 2 \quad \forall i, k_t, t \in \{a, s, c, h, v\}$$

- **Minimum Coverage Requirement:** The total number of personnel assigned to road segment i during shift j must meet or exceed the baseline operational requirement E_{ij} determined by the spatiotemporal forecasting engine:

$$\sum_{k_a=1}^{N_a} x_{ijk_a}^a + \sum_{k_s=1}^{N_s} x_{ijk_s}^s + \sum_{k_c=1}^{N_c} x_{ijk_c}^c + \sum_{k_h=1}^{N_h} x_{ijk_h}^h + \sum_{k_v=1}^{N_v} x_{ijk_v}^v \geq E_{ij} \quad \forall i, j$$

- **Supervision Ratios:** Non-sworn volunteers must operate under the direct oversight of a sworn Constable. The model enforces that the volunteer-to-Constable ratio does not exceed the threshold M :

$$\sum_{k_h=1}^{N_h} x_{ijkh}^h + \sum_{k_v=1}^{N_v} x_{ijkv}^v \leq M \sum_{k_c=1}^{N_c} x_{ijkc}^c \quad \forall i, j$$

In addition to short-term scheduling, long-term manpower planning requires calculating the Relief Factor—the multiplier used to determine the total staff required to support a single 24/7 post.¹⁴ To cover one continuous 8-hour shift year-round requires 2,912 hours of active staffing annually.¹⁴ Although a standard full-time employee is contracted for 2,080 hours per year, training, sick leave, and vacation reduce actual working hours.¹⁴ The Relief Factor (RF) is calculated as:

$$RF = \frac{\text{Required Annual Post Hours}}{\text{Actual Productive Hours worked per Officer}}$$

This ratio typically ranges from 1.6 to 2.1, providing an empirical baseline for budget requests and strategic staffing allocations.¹⁴

For complex urban networks, administrators resolve the allocation problem using lexicographic goal programming.¹¹ This approach establishes a preemptive priority structure among conflicting goals, such as budget limits, minimum coverage targets, and safety metrics¹¹:

$$\text{Min } A = [g_1(n_1, p_1), g_2(n_2, p_2), \dots, g_k(n_k, p_k)]$$

where n_i and p_i represent the negative and positive deviations from target goal i , and the priority hierarchy enforces that higher-level goals must be satisfied before lower-level objectives are evaluated.¹¹

System Optimum Diversion Schemes and Traffic Weighted Multi-Map (TWM) Routing

To effectively clear congestion, routing engines must shift from local user-centric optimization to system-wide capacity balancing.²⁴ Standard navigation applications direct vehicles toward User Equilibrium (UE), where no driver can unilaterally reduce their travel time.⁴³ However, this individualistic routing causes secondary bottlenecks, as multiple drivers are directed onto the same local streets, exceeding their design capacity.²²

Approaching System Optimum (SO) Conditions

The System Optimum condition minimizes the total travel time of all vehicles in the network.²⁴ This can be achieved by replacing standard edge weights with marginal travel times (MTT), representing the incremental delay a vehicle imposes on other road users.⁴³ MTT is estimated through local or global approximations⁴³:

$$\text{MTT}_{e,t} = \tau_{e,t} + V_{e,t} \cdot \frac{\partial \tau_{e,t}}{\partial V_{e,t}}$$

where $\tau_{e,t}$ represents the travel time on segment e , and $V_{e,t}$ is the vehicle volume.⁴³

Traffic Weighted Multi-Maps (TWM) Routing

To achieve system-optimal flow without centralized control, the system utilizes the Traffic Weighted Multi-Maps (TWM) routing strategy.⁴⁶ Rather than distributing a single map representation, the Traffic Management Center generates and distributes M complementary cost maps to different vehicle fleets.⁴⁶ These maps scale segment weights to encourage route diversity⁴⁶:

$$\beta_{e,m} = \alpha_e \cdot S_e \cdot (1 + \delta_m)$$

where S_e represents base segment speed, α_e is a linear scaling factor, and δ_m is a variation factor drawn from normal or uniform distributions.⁴⁷

By calculating alternative routes using semi-disjoint shortest paths with limited spatial overlap, the system spreads the traffic load across the broader network.⁴⁶ Rather than using standard k -shortest paths, which often share 90% of their road segments, the semi-disjoint path algorithm enforces path dissimilarity.⁴⁶ This minimizes corridor overlap and guides vehicles to underutilized segments of the urban network.⁴⁶

The generation of TWM map sets utilizes the ESX-C algorithm to compute these semi-disjoint paths.⁴⁶ The optimal distribution of maps to different vehicle fleets is then solved using genetic evolutionary algorithms (GA) to minimize total network travel time.⁴⁶

Simulations on high-density corridors show that this integrated approach can reduce total travel times by **16.1%** and lower overall exposure to air pollutants by **26.2%** compared to traditional distance-only routing models.⁵⁰

Minimum Viable Product (MVP) Operations and Interactive Dashboard Architecture

The proposed Minimum Viable Product (MVP) is structured around four primary operational dashboards. These screens address the three core gaps identified in traditional traffic management: the lack of predictive impact forecasting, reliance on manual resource deployment, and the absence of post-event learning.⁷

Screen 1: Predictive Event Simulator (Planning Phase)

This screen enables traffic engineers and planners to simulate and evaluate the network-wide footprint of a planned special event.⁴



Screen 1 Narrative

- **Visual Layout and Widgets:** The interface displays input forms on the left and a central

3D GIS map container.⁵¹ The left panel features configuration widgets to set expected attendance, event type, and duration.³¹ Below the map, a line chart plots projected traffic volumes against typical baseline patterns, highlighting the peak demand window.⁴

- **Displayed Metrics and KPIs:** Displays peak inbound traffic volume, maximum projected queue lengths, segment-specific Travel Time Index (TTI) projections, and estimated travel delays.¹
- **Underlying Algorithms:** The simulator utilizes the SUMO microscopic traffic simulation engine.¹ This is coupled with an XGBoost regressor trained on historical event logs to predict the impact on segment-level travel times based on calendar dates and expected attendance.³⁷

Screen 2: Tactical Resource Optimizer (Operational Setup)

This screen converts forecasting results into recommended personnel deployments, physical barricade placement, and corridor-level diversions.⁴¹



Screen 2 Narrative

- **Visual Layout and Widgets:** Features slide-control inputs on the left for adjusting the

optimization parameters of the multi-objective algorithms.¹¹ The recommended deployment schedule displays a detailed table specifying the positioning, shifts, and assigned duties of police sergeants, constables, and civic volunteers.⁴¹ Below this, the map displays the physical control plan, outlining recommended locations for physical barricades, suggested lane diversion angles, and traffic signal override junctions.¹⁰

- **Displayed Metrics and KPIs:** Displays overall operational cost, sergeant-to-risk coverage ratios, civilian auxiliary staff percentages, the calculated Relief Factor for 24/7 post coverage, and required barricade count.¹⁴
- **Underlying Algorithms:** The deployment schedule is computed using a Multi-Objective Linear Program (MOLP) solved via lexicographic priority methods to balance budget constraints against local safety needs.¹¹ The physical barricade layout utilizes the ESX-C semi-disjoint path algorithm to find optimal diversion points.⁴⁶

Screen 3: Live Corridor Control Center (TMC Operations)

This dashboard provides real-time situational awareness, enabling operators to monitor active road conditions and coordinate incident response.⁵²



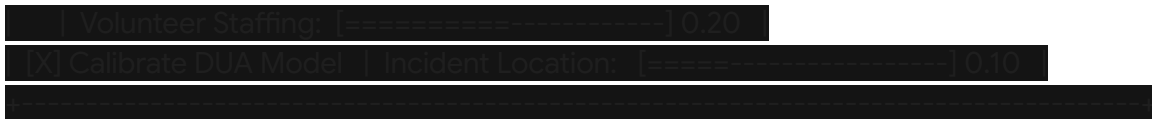
Screen 3 Narrative

- **Visual Layout and Widgets:** Displays active incident alerts on the left, detailing location coordinates, incident class, and estimated clearance times calculated using deep survival analysis.⁵ It integrates a live CCTV camera viewport that updates automatically based on the operator's active map selection.⁵² The central map displays real-time speed fields along primary corridors, colored by delay severity.⁵⁴ Below the map, dynamic corridor metrics show system-wide performance, including travel time indices and overall incident clearance progress.⁵⁴
- **Displayed Metrics and KPIs:** Displays real-time segment speeds, travel time index (TTI), active incident count, responder dispatch times, and estimated incident clearance curves.³⁶
- **Underlying Algorithms:** The incident duration forecaster utilizes the DeepSurv neural network model to estimate clearance time, leveraging historical logs and responder-to-vehicle alert metrics.² The real-time GIS speed map is updated via streaming spatial filtering over GPS vehicle trajectories.⁵²

Screen 4: Post-Event Debriefing & Learning Hub (Closed-Loop Analytics)

This screen provides the tools to conduct after-action reviews, analyze operational bottlenecks, and update predictive parameters.⁸





Screen 4 Narrative

- **Visual Layout and Widgets:** The left panel summarizes the event after-action statistics, comparing target clearance times and queue lengths against actual performance.⁸ The central graph plots projected traffic metrics against actual measurements, highlighting operational mismatches.¹⁶ Below this plot, a SHAP feature importance chart displays the primary variables that drove predictive errors during the event, identifying if issues were caused by incorrect barricade positioning, insufficient volunteering staff, or inaccurate diversion routes.³⁴
- **Displayed Metrics and KPIs:** Displays excess travel delay hours, interagency message counts, variance in signal overrides, the coefficient of determination (R^2) for the forecasting model, and actual-versus-predicted volume deviation percentages.³⁴
- **Underlying Algorithms:** The error diagnosis engine utilizes SHAP explainer networks on the gradient-boosted models to calculate feature attribution for the observed traffic deviations.³⁴ The feedback module updates future simulation parameters using a closed-loop learning framework.¹⁶

Closed-Loop Machine Learning and Post-Event Learning Cycles

To prevent performance degradation over time, the system transitions from static predictions to a closed-loop machine learning model.⁷ Standard predictive systems often suffer from covariate shift when evaluated in real-world environments.⁵⁹ This issue arises because offline models are trained on historical driver trajectories, but the real-time routing recommendations adjust driver paths, altering the underlying traffic distribution.⁴⁶

Mitigating Covariate Shift

The closed-loop architecture addresses covariate shift by evaluating predictive performance against real-world observations.¹⁶ During day-of-event operations, the system samples live network trajectories to compare predicted and actual travel times.¹⁶ If deviations exceed an uncertainty threshold, the system flags the issue and updates the routing model.¹⁶ This process relies on Reinforcing Traffic Rules (RTR) and Closest-Among-Top-K (CAT-K) rollout training.⁵⁹ The system runs closed-loop simulations of both typical and rare "long-tail" scenarios.⁶⁰ Rather than training the models in an open-loop fashion, CAT-K rollouts utilize real-time trajectory updates, enabling lighter, lower-parameter models to outperform larger architectures.⁵⁹ This closed-loop approach reduces model compounding errors, ensuring stable, real-time routing suggestions.⁶⁰



Post-Event Evaluation Framework

The learning cycle concludes with a structured post-event evaluation, aligned with the Federal Highway Administration's (FHWA) special event management standards.⁸ Operational performance is divided into internal and external metrics to assess system efficiency⁵⁷:

- **Internal Performance Metrics:**

- *Roster Efficiency*: Total operational cost deviation from optimized MOLP schedules.⁴¹
- *Infrastructure Adjustments*: Total number of active traffic signal overrides, physical barricades deployed, and messages displayed on dynamic message signs.⁷
- *Communication Volume*: Number of inter-stakeholder messages transmitted between the Traffic Management Center, municipal planners, and field units.⁸

- **External Performance Metrics:**

- *Network Impact*: Total vehicle volume on major and alternative routes.⁵⁷

- *Queue & Delay Performance*: Cumulative hours of travel delay and peak queue lengths.⁷
- *Corridor Throughput*: Travel time indices (TTI), average segment speeds, and modal split (including public transit utilization).¹

By analyzing these metrics during after-action reviews, the system updates future simulation baselines and optimizes resources.¹⁰ This continuous calibration cycle updates the cost matrices of the multi-objective deployment model and refines the spatial parameters of the routing engines, establishing a self-improving spatiotemporal traffic management framework.⁷

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